AMBIENT TEMPERATURE SENSOR DIAGNOSTICS

The outside temperature function is supported by the ambient temperature sensor, a signal and ground circuit hardwired to the compass/temp module, and the Compass/Temp Mirror display.

If the Compass/Temp Mirror display indicates ″SC″, the ambient temperature sensor circuit is shorted to ground.

If the Compass/Temp Mirror display indicates ″OC″, the ambient temperature sensor circuit is open.

The ambient temperature sensor can be diagnosed using the following Sensor Test. First, confirm that °C is not being mistaken for °F or viseversa. If the ambient temperature sensor and the circuits are confirmed to be OK, but the temperature display is inoperative or incorrect, replace the Compass/Temp Mirror.

AMBIENT TEMPERATURE SENSOR TEST

1. Turn the ignition OFF.
2. Disconnect and isolate the battery negative cable.
3. Disconnect the ambient temperature sensor harness connector.
4. Measure the resistance of the ambient temperature sensor using the following values:
   - > 0° C (32° F) Sensor Resistance = 29.33 - 35.99 Kilohms
   - > 10° C (50° F) Sensor Resistance = 17.99 - 21.81 Kilohms
   - > 20° C (68° F) Sensor Resistance = 11.37 - 13.61 Kilohms
   - > 25° C (77° F) Sensor Resistance = 9.12 - 10.86 Kilohms
   - > 30° C (86° F) Sensor Resistance = 7.37 - 8.75 Kilohms
   - > 40° C (104° F) Sensor Resistance = 4.90 - 5.75 Kilohms

The sensor resistance should read between these min/max values. If the resistance value is OK, refer to the Wiring Diagrams to test the Signal and Ground circuits. If the resistance values are not OK, replace the Sensor.

MAP/READING LAMP OPERATION

The Map/Reading lamp feature uses LED (light emitting diode) technology as its light source. The driver and passenger map lamps each consist of 6 LED's, 2 blue-green and 4 amber. By mixing these colors through the use of a diffusing cover the output light is white in appearance. Direct viewing of the LED's will appear as blue-green and amber. The lights are activated by the door switch when entering or exiting the vehicle or manually by using the appropriate button on the mirror as described above. LED's last 10-15+ years and are not designed to be replaced in this application.

3.4 ELECTRICALLY HEATED SYSTEMS

3.4.1 REAR WINDOW DEFOGGER

The timing circuit for the Rear Window Defogger is contained internally within the cluster. When the ignition is on, the first actuation of the Rear Window Defogger switch initiates a 10 (ten) minute time cycle. After 10 (ten) minutes of the same ignition cycle has elapsed, the cluster will turn the defogger off by opening the ground to the Rear Window Defogger Relay. If the defogger switch is actuated a second time during the same ignition cycle, the cluster will turn the defogger off after 5 (five) minutes. While the defogger is on, depressing the switch a second time will turn the defogger off. If the ignition is turned off while the defogger is operating, the defogger will remain off when the ignition is cycled.

3.5 ELECTRO/MECHANICAL INSTRUMENT CLUSTER (EMIC)

3.5.1 SMART CLUSTER FEATURES

The Electro/Mechanical Instrument Cluster (EMIC) houses the Fuel, Voltmeter, Engine Coolant Temp, and Oil Pressure gauges, the Speedometer and the Tachometer. The cluster positions the analog gauges using PCI Bus messages received from the PCM. The cluster also contains warning indicators as well as indicators for the Right and Left turn signals and the High Beam headlamps. Some of the indicators are hardwired to the cluster and some indicators are controlled by messages received on the PCI Bus. The vehicle Chime function, Courtesy Lamp Output, the Battery Saver function, and the Rear Window Defogger timer are contained internally within the cluster. The cluster contains a vacuum fluorescent (VF) display for the Odometer/ Trip function. The cluster VF will also display P-codes and vehicle status messages. The cluster has the ability to store DTCs, communicate on the PCI Bus, display engine information, and display certain inputs using the DRBIII®. The cluster is also able to perform a manual self-test. For complete description and operation of the Instrument Cluster, refer to the TJ Service Manual Instrument Cluster section. For diagnostic procedures, refer to the TJ Body Diagnostic Procedures Manual.